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Roll no.

Degree: B.Tech. Semester: 2 Course work
END-SEMESTER EXAMINATION (SUMMER), July 2026

Course Title: Analog and Digital Electronics
Course Code: EECEC203

Duration: 3:00 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1	Attempt any 2 parts of the following:		
1a	Compare Zener breakdown and avalanche breakdown with respect to the breakdown mechanism, doping concentration, depletion layer width, breakdown voltage, temperature coefficient, and practical applications.	5	1
1b	A bridge rectifier is supplied with an input voltage of 24 V (rms). Assuming ideal diodes, calculate the peak output voltage, average DC output voltage, rectification efficiency, and ripple factor. Compare the results with those of a half-wave rectifier.	5	1
1c	Compare half-wave, center-tapped full-wave, and bridge rectifiers with respect to efficiency, ripple factor, PIV, transformer utilization factor, and applications.	5	1
2	Attempt any 2 parts of the following:		
2a	Compare the Common Base (CB), Common Emitter (CE), and Common Collector (CC) transistor configurations with respect to input impedance, output impedance, current gain, voltage gain, power gain, phase relationship, and applications.	5	2
2b	A silicon transistor operating in the Common Emitter (CE) configuration has $\beta = 100$, base current (I_B) = 20 μ A, $V_{CC} = 15$ V, and collector resistance (R_C) = 2.2 k Ω . Calculate the collector current (I_C), emitter current (I_E), collector-emitter voltage (V_{CE}), and determine whether the transistor operates in the cutoff, active, or saturation region.	5	2
2c	Explain the small-signal analysis of a Common Emitter (CE) amplifier. Derive the expressions for voltage gain, input impedance, and output impedance using the small-signal model.	5	2
3	Attempt any 2 parts of the following:		
3a	An op-amp has a slew rate of 0.8 V/ μ s and a gain-bandwidth product of 2 MHz. Determine whether it can amplify a 100 kHz sinusoidal signal with a peak output voltage of 12 V without distortion.	5	3
3b	Explain the significance of noise margin, propagation delay, fan-in, and fan-out in digital logic families. Discuss how these parameters influence circuit performance.	5	3

3c	Explain the transistor-level implementation of a CMOS NAND gate. Discuss its operation for all possible input combinations.	5	3
4	Attempt any 2 parts of the following:		
4a	Reduce the following Boolean function to its simplest Sum-of-Products (SOP) form using a 5-variable Karnaugh map. $F(A, B, C, D, E) = \sum m(0, 1, 6, 7, 10, 11, 12, 13, 16, 17, 21, 23, 25, 26, 30)$	5	4
4b	Design a 2-bit magnitude comparator using basic logic gates. Derive the Boolean expressions for the outputs $A > B$, $A = B$, and $A < B$, and explain its operation with a suitable truth table.	5	4
4c	Implement the following Boolean function using a 4:1 multiplexer: $F(A, B, C) = \sum m(1, 2, 3, 5, 7)$ (i) Using A and B as the select lines. (ii) Using B and C as the select lines.	5	4
5	Attempt any 2 parts of the following:		
5a	Explain the working of a Ring Counter. Discuss its limitations and describe how a Johnson Counter overcomes them with suitable diagrams.	5	5
5b	What is the race-around condition in a JK flip-flop? Explain how a Master-Slave JK flip-flop eliminates the race-around condition with the help of a neat diagram.	5	5
5c	Derive the excitation tables of the SR and JK flip-flops. Hence, convert a JK flip-flop into an SR flip-flop using suitable logic and explain the design.	5	5